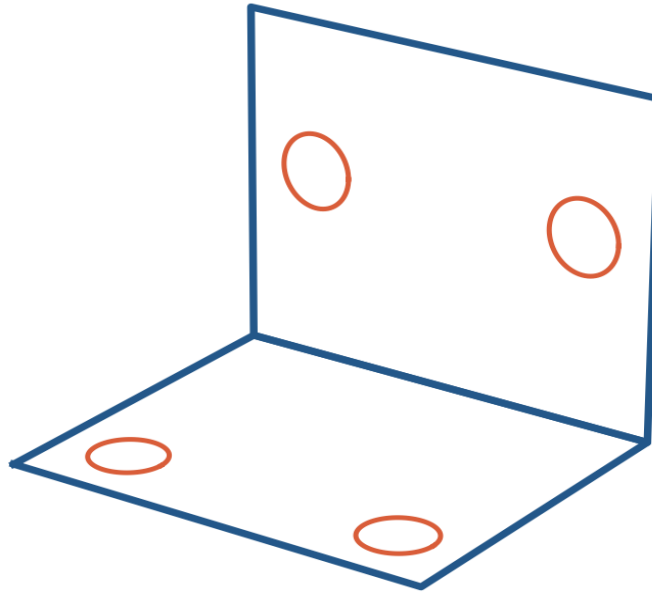


Mechanical Bracket Design and Process Validation

**Coursework-based academic project applying MCE 516
Production Planning and Control concepts**

Conceptual L Bracket with Four Mounting Holes



Conclusion. Hole diameter, hole spacing, and verticality exceed the capability-index target of 1.33 in this simulated five-build analysis. Base width remains below the target and requires tighter forming control before a physical build.

1 Project scope

This academic portfolio project applies concepts from MCE 516 Production Planning and Control at Cleveland State University to mechanical design validation. The bracket case study was developed after completing the course. No physical parts were measured; all inspection values are simulated.

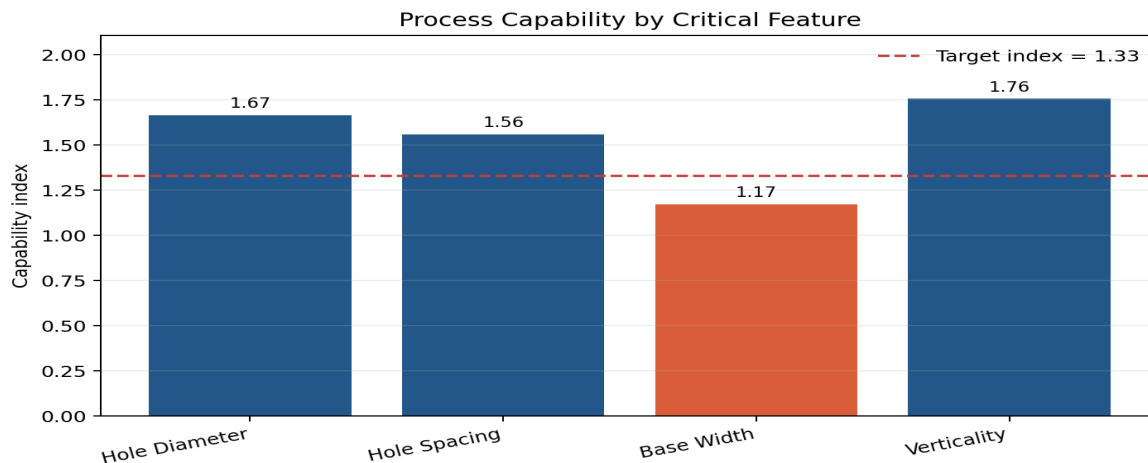
2 Design definition

The part is a 60 mm wide L bracket with a 50 mm base, 45 mm upright, 5 mm thickness, and four 10 mm mounting holes. Aluminum 6061-T6 was selected as a common lightweight structural material. Datum A is the base mounting surface; verticality controls the upright relative to the base.

3 Validation method

Five simulated builds were evaluated with five samples per build. Two-sided dimensions use Cp and Cpk. Verticality has only an upper tolerance limit, so it uses Cpu. All indices use the sample standard deviation across 25 observations, with 1.33 as the portfolio analysis target.

Feature	Nominal	LSL	USL	Mean	Stdev	Index	Value	Status
Hole Diameter	10.00	9.90	10.10	10.013	0.017	Cpk	1.67	PASS
Hole Spacing	40.00	39.85	40.15	40.022	0.027	Cpk	1.56	PASS
Base Width	60.00	59.80	60.20	59.990	0.054	Cpk	1.17	REVIEW
Verticality	0.00	--	0.20	0.078	0.023	Cpu	1.76	PASS



4 Engineering recommendation

Before physical prototyping, review bend allowance, forming-fixture repeatability, fastener access, and load requirements. Prioritize reducing base-width variation and continue monitoring verticality to datum A. The DXF and STL are conceptual, not production geometry. Limitations: the study excludes FEA, physical load testing, cost validation, and measured prototype data; simulated capability results are not production evidence.